Section 5.5 Distributions & repeated trials

分布与重复试验

Method 1 / 方法 1 [PS05-M01]

Distinguish a probability mass from an outcome count. Independent trials give binomial, multinomial, geometric, or state-dependent laws; the right law depends on whether order, stopping, or memory matters.

必须区分点概率与结果数。独立试验可形成二项、多项、几何或状态分布；使用哪种分布，取决于顺序、停止规则及历史记忆。

Pair symmetric score outcomes and count the balance cases

配对对称得分并单独计算平衡情形

Procedure

1. Find a probability-preserving involution exchanging the two strict score comparisons.

2. Count the fixed balance event with the correct multinomial weights.

3. Subtract the balance probability from one and divide by two.

4. Check that the two compared categories have identical laws before using this symmetry.

通用步骤

1. 寻找保持概率且交换两个严格得分比较的对合。

2. 用正确的多项权重计算平衡事件。

3. 从一减去平衡概率，再除以二。

4. 使用该对称前，确认两个被比较类别的规律相同。

> Example. Three independent increments equal $1, -1, 0$ with probabilities $1/4, 1/4, 1/2$. Find the probability their sum is positive.

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 1 / 方法 1 [PS05-M01] (continued / 续)

> 例题：三个独立增量各以概率 $1/4, 1/4, 1/2$ 取 $1, -1, 0$，求总和为正的概率。

Worked solution

1. Negating all increments exchanges positive and negative totals and preserves their probabilities.

2. A zero total is either three zeros, probability $1/8$, or one of each symbol, probability $6(1/4)^2(1/2) = 3/16$. Its total probability is $5/16$.

3. Thus the positive-total probability is $(1 - 5/16)/2 = 11/32$. Counting the three symbols as equiprobable would be incorrect.

例题解析

1. 全部增量反号，会交换正和与负和，且概率不变。

2. 总和为零只能是三个零，概率 $1/8$，或三种符号各一个，概率 $6(1/4)^2(1/2) = 3/16$，合为 $5/16$。

3. 因此正和概率为 $(1 - 5/16)/2 = 11/32$，不能把三个符号当成等概率。

A tie must be removed before a symmetry argument halves the sample space.

利用对称性把样本空间平分之前，必须先扣去平局。

Practice / 配套练习: AIME-2001-II-Q11 (WB p. 789)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 2 / 方法 2 [PS05-M02]

Compare binomial scores by complementing one count

用一个计数的补数比较二项得分

Procedure

1. Write the score inequality in terms of independent remaining-success counts.

2. For fair trials, replace one count Y from m trials by $m-Y$, which has the same binomial law.

3. Use convolution or Vandermonde to turn the comparison into a single binomial tail.

4. Only when the original two laws match may exchange symmetry split the strict comparisons equally.

通用步骤

1. 将得分比较写成两个独立剩余成功次数的不等式。

2. 公平试验中，将 m 次中的 Y 替换为同分布的 $m-Y$。

3. 用卷积或范德蒙德恒等式，把比较转为一个二项尾事件。

4. 只有原来两种分布相同时，交换对称性才能平分严格大小事件。

> Example. Independently let X count heads in three fair flips and Y heads in two fair flips. Find $\Pr(X \geq Y)$.

> 例题：独立地，X 为三次公平投币正面数，Y 为两次正面数，求 $\Pr(X \geq Y)$。

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 2 / 方法 2 [PS05-M02] (continued / 续)

Worked solution

1. The variable $2-Y$ is an independent two-flip head count. Thus $Z=X+(2-Y)$ is a five-flip head count.

2. The event $X \geq Y$ is exactly $Z \geq 2$.

3. Its probability is $1-(\binom{5}{0}+\binom{5}{1})/2^5=26/32=13/16$. The unequal trial counts prevent directly halving the non-tie event.

例题解析

1. $2-Y$ 仍为独立的两次公平投币正面数，所以 $Z=X+(2-Y)$ 为五次正面数。

2. 事件 $X \geq Y$ 等价于 $Z \geq 2$。

3. 概率为 $1-(\binom{5}{0}+\binom{5}{1})/2^5=13/16$。两边试验次数不同，不能直接把非平局事件平分。

The convolution identity $\sum_j \binom{n}{j}\binom{m}{j}=\binom{n+m}{n}$ counts equal scores.

卷积恒等式 $\sum_j \binom{n}{j}\binom{m}{j}=\binom{n+m}{n}$ 可计算相等得分。

Practice / 配套练习: AIME-2006-II-Q10 (WB p. 785)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 3 / 方法 3 [PS05-M03]

Enumerate multinomial count vectors under a weighted total

按加权总量枚举多项计数向量

Procedure

1. Let n_i count occurrences of each outcome value. Write both $\sum n_i = n$ and the weighted-sum equation.

2. Bound and list all nonnegative integer count vectors satisfying the equations.

3. A vector has $n!/\prod n_i!$ orders and probability $(n!/\prod n_i!)\prod p_i^{n_i}$.

4. Sum the disjoint vector probabilities; a common denominator is valid only when the elementary outcomes really are equiprobable.

通用步骤

1. 令 n_i 为各结果出现次数，同时写出总次数及加权总和方程。

2. 给出界并列全满足条件的非负整数计数向量。

3. 一个向量有 $n!/\prod n_i!$ 种顺序，概率为其乘 $\prod p_i^{n_i}$。

4. 相加互斥向量的概率；只有基本结果确实等可能，才能统一按结果数作分母。

> Example. Four independent draws take values $0, 1, 3$ with probabilities $1/2, 1/3, 1/6$. Find the probability their sum is three.

> 例题：四次独立抽取，各以概率 $1/2, 1/3, 1/6$ 取值 $0, 1, 3$，求总和为三的概率。

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 3 / 方法 3 [PS05-M03] (continued / 续)

Worked solution

1. If a three occurs, the other values are all zero, giving counts $(3,0,1)$ and probability $4(1/2)^3(1/6)=1/12$.

2. Without a three, there must be three ones and one zero, giving probability $4(1/3)^3(1/2)=2/27$.

3. The cases are disjoint, so the total is $1/12+2/27=17/108$.

例题解析

1. 若出现三，其余都为零，次数向量为 $(3,0,1)$，概率 $4(1/2)^3(1/6)=1/12$。

2. 若不出现三，就需三个一和一个零，概率为 $4(1/3)^3(1/2)=2/27$。

3. 两类互斥，总概率为 $1/12+2/27=17/108$。

A fixed total can arise from several count vectors; one convenient vector is not an exhaustive list.

固定总和可能对应多个计数向量，找到一个方便向量并不等于穷尽。

Practice / 配套练习: AIME-2008-I-Q09 (WB p. 786)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 4 / 方法 4 [PS05-M04]

Cancel binomial probabilities to recover success odds

约去二项概率以恢复成功赔率

Procedure

1. Write both exact-success probabilities with their binomial coefficients.

2. Divide by positive common factors, preserving any unequal combinatorial coefficient.

3. Solve for the positive odds $p/(1-p)$ and restore $0 < p < 1$.

4. Check the original probability ratio rather than only the powered equation.

通用步骤

1. 写出两个准确成功次数的二项概率及组合系数。

2. 除去正的公共因子，但保留不相等的组合系数。

3. 求正赔率 $p/(1-p)$，并恢复 $0 < p < 1$。

4. 核对原概率比，而不只核对取幂后的方程。

> Example. In four independent flips of a coin with head probability $0 < p < 1$, zero heads and exactly two heads have equal probabilities. Find p.

> 例题：独立投一枚正面概率满足 $0 < p < 1$ 的硬币四次，零次正面与恰两次正面的概率相等，求 p。

Worked solution

1. The equality is $(1-p)^4 = 6p^2(1-p)^2$. A valid nondegenerate coin has $0 < p < 1$.

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 4 / 方法 4 [PS05-M04] (continued / 续)

2. Cancel $(1-p)^2$ and take the positive square root: $1-p=\sqrt{6}p$.

3. Thus $p=1/(1+\sqrt{6})=(\sqrt{6}-1)/5$, which lies in $(0,1)$ and satisfies the original equality.

例题解析

1. 方程为 $(1-p)^4=6p^2(1-p)^2$，非退化硬币满足 $0<p<1$。

2. 约去 $(1-p)^2$ 并取正平方根，得 $1-p=\sqrt{6}p$。

3. 故 $p=(\sqrt{6}-1)/5$，位于零与一之间，代回原式成立。

Equal binomial coefficients are a special cancellation, not a general rule.

两边组合系数相等只是特殊情况，不能普遍略去。

Practice / 配套练习: AIME-2009-I-Q03 (WB p. 783)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 5 / 方法 5 [PS05-M05]

Count allowed prefixes before attaching trial weights

先数合法前缀再附加试验权重

Procedure

1. Let $D(n, j)$ count allowed prefixes of length n with j successes.

2. Update by success or failure, discarding each state that violates the prefix boundary at that time.

3. If all final words have the same success total, multiply their count by the common Bernoulli word weight.

4. Keep the terminal rule separate when the boundary applies only to proper prefixes.

通用步骤

1. 令 $D(n, j)$ 为长 n、含 j 次成功的合法前缀数。

2. 按成功或失败转移，每一步都丢弃当时越过前缀边界的状态。

3. 若终态成功次数相同，各词权重相同，可用词数乘共同伯努利权重。

4. 若边界只限制真前缀，须单独处理末步规则。

> Example. Six independent trials succeed with probability $1/3$. Find the probability of exactly two successes with no prefix having more successes than failures.

> 例题：六次独立试验成功概率为 $1/3$，求恰两次成功且任一前缀中成功数不多于失败数的概率。

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 5 / 方法 5 [PS05-M05] (continued / 续)

Worked solution

1. Use $D(0,0)=1$ and $D(n,j)=D(n-1,j)+D(n-1,j-1)$ only when $2j \le n$.

2. The rows through $n=6$, restricted to relevant successes, end with $D(4,2)=2, D(5,2)=5, D(6,2)=9$; the last follows $D(5,1)+D(5,2)=4+5$.

3. Each accepted word has weight $(1/3)^2(2/3)^4$. Thus the probability is $9 \cdot 16/729 = 16/81$.

例题解析

1. 取 $D(0,0)=1$，仅保留 $2j \le n$ 的状态，用递推 $D(n,j)=D(n-1,j)+D(n-1,j-1)$。

2. 逐行计算得 $D(4,2)=2, D(5,2)=5, D(6,2)=9$，末项来自 $D(5,1)+D(5,2)=4+5$。

3. 每个合法词权重为 $(1/3)^2(2/3)^4$，所以概率为 $9 \cdot 16/729 = 16/81$。

A correct final success fraction does not guarantee that every earlier prefix was legal.

最终成功比例正确，不代表每个先前前缀都合法。

Practice / 配套练习: AIME-2002-II-Q12 (WB p. 790)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 6 / 方法 6 [PS05-M06]

Match independent stopping counts by summing mass products

累加点概率乘积以匹配独立停止次数

Procedure

1. Write each complete stopping-count mass, such as $p(1-p)^{k-1}$.

2. For equal counts sum $\sum_k p_k q_k$; for an offset sum the shifted products over their common support.

3. Use a geometric series for infinite support and include both offset directions separately.

4. Do not double one offset when the two distributions differ.

通用步骤

1. 写出完整停止次数的点概率，如 $p(1-p)^{k-1}$。

2. 次数相等时求 $\sum_k p_k q_k$，有偏移时按共同支持集累加移位乘积。

3. 无穷支持用等比级数，两个偏移方向要分别计入。

4. 若两种分布不同，不能把某一方向直接乘二。

> Example. Two independent processes stop on their first successes, with per-trial success probabilities $1/2$ and $1/3$. Find the chance their stopping counts differ by at most one.

> 例题：两个独立过程分别以每次成功概率 $1/2$、$1/3$ 等待首次成功，求停止次数相差不超过一的概率。

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 6 / 方法 6 [PS05-M06] (continued / 续)

Worked solution

1. Equal counts have probability $\sum_{k\ge1}(1/2)^k(1/3)(2/3)^{k-1}=(1/6)/(1-1/3)=1/4$.

2. The first process stopping one later contributes $(1/2)(1/4)=1/8$; the second stopping one later contributes $(2/3)(1/4)=1/6$.

3. These are disjoint, totaling $1/4+1/8+1/6=13/24$. Their unequal offset weights show why symmetry cannot be assumed.

例题解析

1. 次数相等概率为 $\sum_{k\ge1}(1/2)^k(1/3)(2/3)^{k-1}=1/4$。

2. 第一个过程晚一次的概率为 $(1/2)(1/4)=1/8$，第二个晚一次为 $(2/3)(1/4)=1/6$。

3. 三类互斥，总概率为 $1/4+1/8+1/6=13/24$。两方向权重不同，不能假设对称。

Equality of two independent counts is a sum of products of masses, not a product of their total probabilities.

两个独立次数相等应累加各点概率的乘积，而不是乘各自总概率。

Practice / 配套练习: AIME-2009-II-Q08 (WB p. 787)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 7 / 方法 7 [PS05-M07]

Multiply probability-generating polynomials for unequal trials

用概率生成多项式处理不同成功率

Procedure

1. A Bernoulli trial with success probability p contributes $(1-p)+pt$.

2. Multiply these factors for independent trials, retaining exact coefficients.

3. The coefficient of t^j is the full probability mass for j successes.

4. For independent repetitions, match masses by a sum of squared coefficients or convolve the two different laws.

通用步骤

1. 成功概率 p 的伯努利试验对应 $(1-p)+pt$。

2. 独立试验对应因子的乘积，保留精确系数。

3. t^j 的系数就是成功 j 次的完整点概率。

4. 独立重复时，以系数平方和求同次数概率；两种不同分布则作对应卷积。

> Example. Three independent coins have head probabilities $1/2, 1/3, 2/3$. Find the probability of exactly two heads.

> 例题：三枚独立硬币的正面概率为 $1/2, 1/3, 2/3$，求恰两次正面的概率。

Worked solution

1. The generating polynomial is $((1+t)/2)((2+t)/3)((1+2t)/3)$.

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分布与重复试验

Method 7 / 方法 7 [PS05-M07] (continued / 续)

2. Expanding gives $(2+7t+7t^2+2t^3)/18$; its coefficients sum to one.

3. The coefficient of t^2 is $7/18$, the desired probability. Equal-probability binomial formulas would not apply to these three coins.

例题解析

1. 生成多项式为 $((1+t)/2)((2+t)/3)((1+2t)/3)$。

2. 展开为 $(2+7t+7t^2+2t^3)/18$，系数和为一。

3. t^2 系数为 $7/18$，即所求概率；三枚硬币成功率不同，不能套同参数二项公式。

Check normalization at $t=1$ before using any coefficient as a probability.

把系数当作概率之前，先代入 $t=1$ 检查总和为一。

Practice / 配套练习: AIME-2010-I-Q04 (WB p. 784)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 8 / 方法 8 [PS05-M08]

Propagate a walk distribution with enough memory

以足够历史信息传播行走分布

Procedure

1. Choose a state recording all information needed for the next move; a no-backtracking rule needs the previous edge or equivalent type.

2. Group states only when their class transition totals match.

3. Use either probabilities or path counts consistently; if using counts, prove that every complete path has the same weight.

4. Initialize after any exceptional first move and iterate to the requested horizon.

通用步骤

1. 状态须记录决定下一步的全部信息；禁止立即返回时，要记上一条边或等价类型。

2. 只有各类别转移总量相同时才能合并状态。

3. 统一使用概率或路径数；若数路径，必须证明完整路径等权。

4. 首步若规则特殊，应在首步后设初值，再迭代到所求时刻。

> Example. A walker on the complete graph with four vertices starts at A. Its first move is uniform among three neighbors; later moves are uniform among neighbors other than the previous vertex. Find its return probability after four moves.

> 例题：行者在四顶点完全图上从 A 出发，首步等可能去三个邻点，之后等可能去除上一顶点外的邻点。求四步后回到 A 的概率。

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 8 / 方法 8 [PS05-M08] (continued / 续)

Worked solution

1. After two moves the path is A, B, C with all three distinct; symmetry makes the actual names irrelevant.

2. The third step goes to A or the fourth vertex D, each with probability $1/2$. From A at step three, step four cannot remain at A.

3. From D at step three, excluding C leaves A and B as equiprobable choices. Thus the four-step return probability is $(1/2)(1/2) = 1/4$.

例题解析

1. 两步后的路径为互异三点 A, B, C，由对称性，具体标号无关。

2. 第三步等可能到 A 或第四点 D；若第三步到 A，第四步不可能留在 A。

3. 若第三步到 D，排除上一点 C 后，可等可能去 A 或 B。所以四步返回概率为 $1/4$。

Being at the same vertex does not imply the same future law when the incoming edge matters.

若来边影响可走方向，到达同一顶点并不代表未来规律相同。

Practice / 配套练习: AIME-2003-II-Q13 (WB p. 791), AIME-2021-II-Q08 (WB p. 788)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 9 / 方法 9 [PS05-M09]

Test independence after a coordinate transformation

坐标变换后重新检验独立性

Procedure

1. List the transformed increment pair for every elementary move.

2. Independence requires joint probabilities to equal products of marginal probabilities; a symmetric drawing alone is insufficient.

3. If every sign pair occurs with probability $1/4$, the two coordinate walks are independent fair sign walks.

4. Translate the target event exactly, then use binomial tails and any required union correction.

通用步骤

1. 逐一列出每个基本移动对应的变换后增量对。

2. 独立要求联合概率等于边缘概率乘积，仅有图形对称还不够。

3. 若四种符号对各以概率 $1/4$ 出现，两坐标才是独立的公平正负行走。

4. 准确转换目标事件，再用二项尾概率及必要的并集修正。

> Example. One axis step chooses east or west with probability $1/3$ each, north or south with $1/6$ each. Are $U = x + y$ and $V = x - y$ independent?

> 例题：一步在东西方向各以概率 $1/3$、南北各以 $1/6$ 移动。$U = x + y, V = x - y$ 是否独立?

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 9 / 方法 9 [PS05-M09] (continued / 续)

Worked solution

1. The four transformed pairs for east, west, north, south are $(1,1),(-1,-1),(1,-1),(-1,1)$.

2. Both marginals satisfy $\Pr(U=1)=\Pr(V=1)=1/2$. But $\Pr(U=1,V=1)=1/3$.

3. Since $1/3 \neq (1/2)(1/2)$, the transformed variables are not independent. Equal positive and negative marginals do not certify independence.

例题解析

1. 东西南北对应的变换对分别为 $(1,1),(-1,-1),(1,-1),(-1,1)$。

2. 边缘概率均有 $\Pr(U=1)=\Pr(V=1)=1/2$，但联合概率 $\Pr(U=1,V=1)=1/3$。

3. 由于 $1/3 \neq 1/4$，两变量不独立。边缘分布的正负对称并不能证明独立。

Under uniform axis moves the sign-pair table is uniform; changing move weights can destroy the factorization.

等可能轴向移动给出均匀符号对；改变移动权重可能破坏分解。

Practice / 配套练习: AIME-2014-I-Q11 (WB p. 792)